February 3, 2026

Another 3-D geometry problem... Distance between two lines in $\mathbb{R}^3$

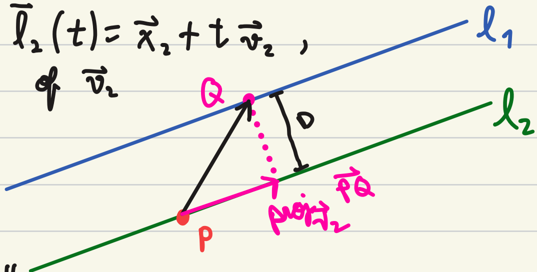
Three cases: 1) Intersecting lines 2) Parallel lines 3) Skew lines

1) Intersecting lines $\vec{l}_1(t)$, $\vec{l}_2(t)$: distance $= 0$

Need to check $\vec{l}_1(t_1) = \vec{l}_2(t_2)$ for some $t_1, t_2 \in \mathbb{R}$.

2) Parallel lines:

Whenever $\vec{l}_1(t) = \vec{x}_1 + t\vec{v}_1$ $\vec{l}_2(t) = \vec{x}_2 + t\vec{v}_2$, and $\vec{v}_1$ is a scalar multiple of $\vec{v}_2$



We can treat this like a problem of distance from a point to a line:

$D = \| \vec{PQ} - \text{proj}_{\vec{v}_2} \vec{PQ} \|$ or $\dfrac{\|\vec{v}_2 \times \vec{PQ}\|}{\|\vec{v}_2\|} = D$

3) Skew lines:

def: We say two lines (l_1, l_2) are skew if they are neither intersecting nor parallel.

Whenever $\vec{l}_1(t) = \vec{x}_1 + t\vec{v}_1$ and $\vec{l}_2(t) = \vec{x}_2 + t\vec{v}_2$ are skew they lie in parallel planes (namely $\Pi_1 : \vec{r}_1(s,t) = \vec{x}_1 + s\vec{v}_1 + t\vec{v}_2$ and $\Pi_2 : \vec{r}_2(s,t) = \vec{x}_2 + s\vec{v}_1 + t\vec{v}_2$) so the distance between the lines is the same as the distance between the two parallel planes

e.g.

1.6. Some n-dimensional geometry

Vectors in $\mathbb{R}^n$ are ordered n-tuples of real numbers

e.g. $(2,1,4,5,3) \in \mathbb{R}^5$, $(1,0,1,0,0,1,0,0,1) \in \mathbb{R}^9$

Operations of vector addition and scalar multiplication work the same (componentwise)

e.g. $(2,1,4,5,3) - 2(3,0,1,-1,0) = (2,1,4,5,3) + (-6,0,-2,2,0)$
$= (-4,1,2,7,3)$

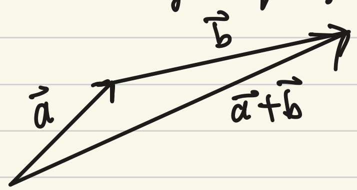
Dot product in $\mathbb{R}^n$: $\vec{a} = (a_1, \dots, a_n)$ $\vec{b} = (b_1, \dots, b_n)$

$$\vec{a} \cdot \vec{b} = a_1 b_1 + \dots + a_n b_n = \sum_{i=1}^{n} a_i b_i \qquad \vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos\theta$$

Proposition (Cauchy-Schwarz inequality): $|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\| \|\vec{b}\|$
for any $\vec{a}, \vec{b} \in \mathbb{R}^n$.

* If you are interested in general vector spaces (possibly infinite dimensional), you should notice Cauchy-Schwarz works every time a "dot product" is positive semi-definite (i.e. $\vec{a} \cdot \vec{a} \geq 0$) and linear (i.e. distributive laws)

Proposition (triangle inequality): $\|\vec{a} + \vec{b}\| \leq \|\vec{a}\| + \|\vec{b}\|$ for any $\vec{a}, \vec{b} \in \mathbb{R}^n$.



Standard basis vectors in $\mathbb{R}^n$:

$$\hat{e}_1 = \vec{e}_1 = (1,0,0,\dots,0)$$
$$\hat{e}_2 = \vec{e}_2 = (0,1,0,\dots,0)$$
$$\vdots$$
$$\hat{e}_n = \vec{e}_n = (0,0,\dots,0,1)$$

we hat to emphasize these are unit vectors, i.e. $\|\hat{e}_i\| = 1$

$$\vec{a} = (a_1, a_2, \dots, a_n) = a_1 \hat{e}_1 + \dots + a_n \hat{e}_n = \sum_{i=1}^{n} a_i \hat{e}_i$$

eg: $(2,0,3) = 2\hat{\imath} + 3\hat{k} = 2\hat{e}_1 + 3\hat{e}_3$

Matrices in $M_{m\times n}(\mathbb{R})$ are rectangular arrays of numbers in $\mathbb{R}$ with m rows and n columns.

Examples and notation :

row number
column number

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = (a_{ij}) \in M_{m\times n}(\mathbb{R}) \qquad a_{ij}$$

e.g. $A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 1 \end{bmatrix}$

$a_{13} = 4$

We can also think of a matrix as a vector of vectors in two ways:

$$A = \begin{bmatrix} (a_{11} & a_{12} & \cdots & a_{1n}) \\ (a_{21} & a_{22} & \cdots & a_{2n}) \\ \vdots & \vdots & & \vdots \\ (a_{m1} & a_{m2} & \cdots & a_{mn}) \end{bmatrix} = \begin{bmatrix} \vec{b}_1 \\ \vdots \\ \vdots \\ \vec{b}_m \end{bmatrix}$$

m row vectors in $\mathbb{R}^n$

$$A = \left[\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} \cdots \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} \right] = [\vec{c}_1 \cdots \vec{c}_n]$$

n column vectors in $\mathbb{R}^m$

eg: $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} \hat{i} \quad \hat{j} \quad \hat{k} \\ \vec{a} \\ \vec{b} \end{vmatrix}$

Operations with matrices (addition, scalar multiplication, and matrix multiplication):

- scalar multiplication : component-wise e.g. $4\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 12 & 4 \\ 0 & 8 \end{bmatrix}$

- Matrix addition : component-wise and only when dimensions match

e.g. $\begin{bmatrix} 1 & -1 \\ 3 & 1 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} 5 & 4 \\ -1 & 6 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 2 & -1 \end{bmatrix}$

properties of matrix addition : For all $A, B, C \in M_{m\times n}(\mathbb{R})$

1) $A+B = B+A$ (commutativity)
2) $A+(B+C) = (A+B)+C$ (associativity)
3) there is $0 \in M_{m\times n}(\mathbb{R})$ (zero matrix) such that $D+0=D$ for all $D \in M_{m\times n}(\mathbb{R})$

properties of scalar multiplication : For all $A, B \in M_{m\times n}(\mathbb{R})$ and $k, l \in \mathbb{R}$

1) $(k+l)A = kA + lA$ (distributivity)
2) $k(A+B) = kA + kB$ (distributivity)
3) $k(lA) = (kl)A = l(kA)$

• Matrix multiplication

Let $A \in M_{m\times n}(\mathbb{R})$ and $B \in M_{n\times p}(\mathbb{R})$, then $AB \in M_{m\times p}(\mathbb{R})$

and the i-j entry in AB is $(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ [a_{i1} & a_{i2} & \cdots & a_{in}] \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1p} \\ b_{21} & \cdots & b_{2j} & \cdots & b_{2p} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{np} \end{bmatrix}$$

$$\parallel$$

$$\begin{bmatrix} \vec{a_1} \\ \vec{a_2} \\ \vdots \\ \vec{a_m} \end{bmatrix} \begin{bmatrix} \vec{b_1} & \cdots & \vec{b_p} \end{bmatrix}$$

$$(AB)_{ij} = \vec{a_i} \cdot \vec{b_j}$$

eg: $\begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 3 & 1 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 6 & -7 \\ 7 & 8 \end{bmatrix}$

properties of matrix multiplication:

1) If $A \in M_{m\times n}$, $B \in M_{n\times p}$, $C \in M_{p\times q}$, then
$(AB)C = A(BC) = ABC$ (associativity)
2) If $A \in M_{m\times n}$, $B \in M_{n\times p}$ and $k \in \mathbb{R}$, then
$k(AB) = (kA)B = A(kB)$
3) If $A, B \in M_{m\times n}$ and $C \in M_{n\times p}$, then
$(A+B)C = AC + BC$ (distributivity)
4) If $A \in M_{m\times n}$ and $B, C \in M_{n\times p}$, then
$A(B+C) = AB + AC$ (distributivity)

def: The transpose of the matrix $A \in M_{m\times n}(\mathbb{R})$ is $A^T \in M_{n\times m}(\mathbb{R})$ and has entries
$$(A^T)_{ij} = A_{ji}$$

e.g. $A = \begin{bmatrix} 2 & 3 & 5 \\ 0 & 1 & 2 \end{bmatrix} \rightsquigarrow A^T = \begin{bmatrix} 2 & 0 \\ 3 & 1 \\ 5 & 2 \end{bmatrix}$

property: If $A \in M_{m\times n}$ and $B \in M_{n\times p}$, then
$(AB)^T = B^T A^T$

proposition: If $A, B \in M_{m\times m}(\mathbb{R})$, then $\det(AB) = \det A \cdot \det B$.

e.g (section 1.6 exercise 14)

14. To make some extra money, you decide to print four types of silk-screened T-shirts that you sell at various prices. You have an inventory of 20 shirts that you can sell for $\$8$ each, 30 shirts that you sell for $\$10$ each, 24 shirts that you sell for $\$12$ each, and 20 shirts that you sell for $\$15$ each. A friend of yours runs a side business selling embroidered baseball caps and has an inventory of 30 caps that can be sold for $\$10$ each, 16 caps that can be sold for $\$10$ each, 20 caps that can be sold for $\$12$ each, and 28 caps that can be sold for $\$15$ each. You suggest swapping half your inventory of each type of T-shirt for half his inventory of each type of baseball cap. Is your friend likely to accept your offer? Why or why not? No, the friend would lose revenue with the swap

Before deal:

revenue of friend

amounts of cap

$$\begin{array}{cccccc} & & \text{Type A} & \text{Type B} & C & D \\ \vec{a} & = & (30, & 16, & 20, & 28) \end{array}$$

price

$$\begin{array}{cccccc} & & A & & & \\ \vec{b} & = & (10, & 10, & 12, & 15) \end{array}$$

$$\vec{a} \cdot \vec{b} = 300 + 160 + 240 + 420$$

$$= 1{,}120 \quad \text{revenue}$$

After deal

revenue of friend

matrix of inventory

$$\begin{array}{ccccccccc} & \text{shirt A} & \text{shirt B} & \text{shirt C} & \text{shirt D} & \text{cap A} & \text{cap B} & \text{cap C} & \text{cap D} \\ A = & [10 & 15 & 12 & 10 & 15 & 8 & 10 & 14] \end{array}$$

mij

$$B = [8 \quad 10 \quad 12 \quad 15 \quad 10 \quad 10 \quad 12 \quad 15]$$

$$\text{revenue} = AB^T = \text{"}\vec{A} \cdot \vec{B}\text{"} = 1084$$